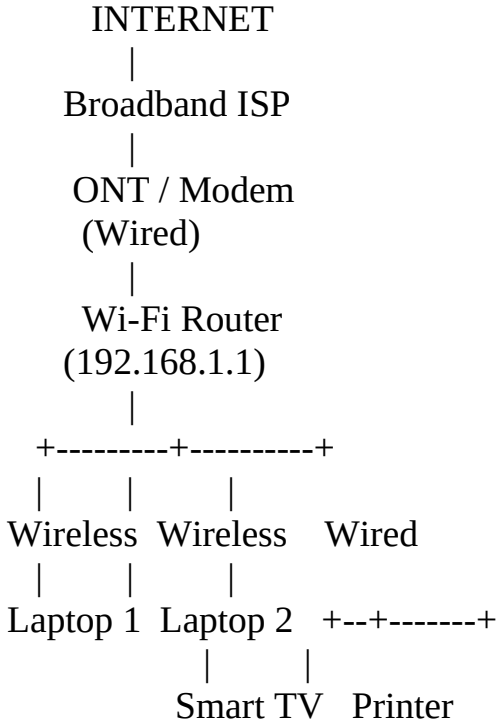
TASK: 1

Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

A SOHO network connects multiple devices to the internet using a Wi-Fi router. The broadband connection provides internet access, and the router distributes the connection to all devices.

Network Diagram:



Connection Types:

1. Broadband to modem: Wired (Optical Fibre).
2. Modem to Wi-Fi router: Wired (Ethernet).
3. Router to Laptop 1: Wireless (Wi-Fi).
4. Router to Laptop 2: Wireless (Wi-Fi).
5. Router to Smart TV: Wired (Ethernet).
6. Router to Printer: Wired (Ethernet).

This network allows all five devices to share the same internet connection.

TASK: 2.

**List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.**

Sr. No.	Hardware Component	Suggested Brand/Model	Purpose
1	Wi-Fi Router	TP-Link Archer C6 AC1200	Connects all devices to the network.
2	Broadband Modem/ONT	ISP-provided ONT	Receives internet from the service provider.
3	Laptop 1	HP Laptop	Connects to the network wirelessly.
4	Laptop 2	Dell Laptop	Connects to the network wirelessly.
5	Smart TV	Samsung Smart TV	Provides internet access for streaming.
6	Network Printer	HP LaserJet Pro	Allows wireless network users to print.
7	Ethernet Cable	D-Link Cat6 Cable	Connects the modem, TV and printer.
8	UPS	APC 600VA UPS	Provides backup power during electricity failures.

Additional Components:

- RJ45 connectors (if making custom Ethernet cables).
- Power adapters for the router and ONT.
- Wi-Fi range extender (optional, if the Wi-Fi signal is weak).
- Network switch (optional, if additional wired devices are required).

TASK: 3.

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

The following steps can be used to configure the TP-Link Archer C6 Wi-Fi router.

Step 1: Connect the Router

- Connect the broadband modem/ONT to the WAN port of the router using an Ethernet cable.
- Switch on the router and wait for the power and internet indicators.

Step 2: Open the Router Admin Page

- Connect a laptop to the router.
- Open a web browser.
- Enter `192.168.1.1` in the address bar.
- Enter the administrator login credentials.

Step 3: Change the Admin Password

- Open System Tools or Administration settings.
- Select Password or Administrator Settings.
- Change the default admin password to a strong, unique password.
- Save the changes.

Step 4: Create a Unique SSID

- Open Wireless Settings.
- Enter a unique Wi-Fi network name (SSID).

Example: `SOHO_Home_Network`

Step 5: Enable WPA2 Security

- Open Wireless Security settings.
- Select WPA2-PSK (Personal).
- Select AES encryption.
- Set a strong Wi-Fi password.
- Save the settings.

Step 6: Enable DHCP

- Open DHCP Server settings.
- Enable the DHCP server.
- Set the IP address range from `192.168.1.100` to `192.168.1.199`.
- Save the configuration.

Step 7: Connect All Devices

- Connect both laptops using Wi-Fi.
- Connect the smart TV and printer using Ethernet cables.
- Enter the Wi-Fi password wherever required.

Step 8: Test the Internet Connection

Open a browser on both laptops and visit a website to check whether the internet is working.

TASK: 4

Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

For this SOHO network, the router will use the IP address `192.168.1.1`.

The printer and smart TV will have static IP addresses, while both laptops will receive IP addresses automatically through DHCP.

Network Configuration:

- Network Address: `192.168.1.0/24`
- Subnet Mask: `255.255.255.0`
- Default Gateway: `192.168.1.1`
- DNS Server: `8.8.8.8`
- DHCP Range: `192.168.1.100 – 192.168.1.199`

IP Address Assignment Table:

Device	MAC Address (Sample)	IP Address	Assignment Type
Wi-Fi Router	02:11:22:33:44:01	192.168.1.1	Static
Printer	02:11:22:33:44:02	192.168.1.10	Static
Smart TV	02:11:22:33:44:03	192.168.1.20	Static
Laptop 1	02:11:22:33:44:04	192.168.1.100	DHCP
Laptop 2	02:11:22:33:44:05	192.168.1.101	DHCP

Configuration Steps:

1. Open the router administration page.
2. Configure the printer IP as `192.168.1.10`.
3. Configure the smart TV IP as `192.168.1.20`.
4. Keep DHCP enabled on the router.
5. Set both laptops to obtain an IP address automatically.
6. Check the assigned IP addresses using the router's DHCP client list or the `ipconfig` command.

TASK: 5.

Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

If one laptop cannot connect to the internet, I would check the following possible causes and solutions.

1. Incorrect Wi-Fi Password

Problem: The laptop may be using an incorrect Wi-Fi password.

Solution:

- Open Wi-Fi settings on the laptop.
- Forget the saved network.
- Select the correct SSID.
- Enter the correct Wi-Fi password.
- Try connecting again.

2. Incorrect IP Address Configuration

Problem: The laptop may not be receiving a valid IP address from the DHCP server.

Solution:

- Open Command Prompt.
- Type `ipconfig` and press Enter.
- Check the IP address and default gateway.
- If necessary, type `ipconfig /release`.
- Then type `ipconfig /renew`.
- Check whether the laptop receives a valid IP address.

3. Weak Wi-Fi Signal

Problem: The laptop may be too far away from the Wi-Fi router.

Solution:

- Move the laptop closer to the router.

- Check the Wi-Fi signal strength.
- Restart the router if necessary.
- Use a Wi-Fi extender if the signal is consistently weak.

4. Router or Internet Connection Problem

Problem: The router or broadband connection may not be working properly.

Solution:

- Check the router's power and internet indicators.
- Restart the router and modem.
- Check whether other devices can access the internet.
- Contact the internet service provider if the connection is still unavailable.

5. Network Driver Problem

Problem: The laptop's Wi-Fi driver may be outdated or corrupted.

Solution:

- Open Device Manager.
- Expand Network Adapters.
- Find the wireless network adapter.
- Update the driver.
- Restart the laptop and try connecting again.